The Salton Sea's Effect on Health

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Roadmap

Introduction	What is the Salton Sea and Why is is important?
Data and Findings	What we are looking at, where we extracted, and what do they show us?
Results	The dynamics between the data and why it makes sense
Policy Review	What should be done going forward

01

Introduction

What is the Salton Sea?

History of the Salton Sea

- Formed by accident in 1905
- Characterized as a terminal lake → water only leaves through evaporation
- 2003 Quantification Settlement Agreement diverted agricultural water away
- As the shoreline recedes, it releases toxic dust

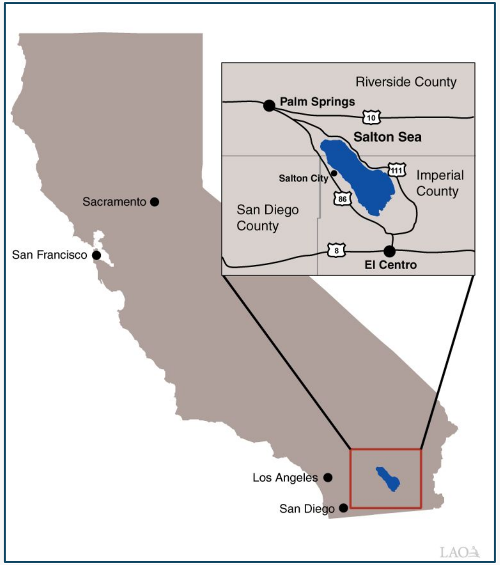

Photo Source: Legislative Analyst's Office

Sources: (Levers et al., 2020), California State Portal

Background on PM10 Air Pollutants

What it is
Particulate matter (PM10) is composed of various airborne particles (e.g., dust, pollen, bacteria fragments) ≤ 10 microns in diameter.

Why it's dangerous
These particles easily bypass the nose and throat, depositing on surfaces of the larger airway region of the lung.

Health Impacts
Exposure can cause worsening of respiratory disease (e.g., asthma, chronic obstructive pulmonary disease)

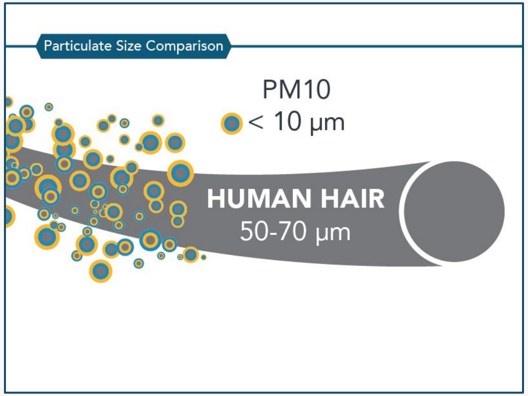

Photo Source: California Air Resources Board

Sources: Environmental Protection Agency (EPA), California Air Resources Board (CARB)

Who Bears the Burden?

Imperial County, CA

87% Hispanic Population	$60,749 Median Household Income (U.S. med: $83,730)	19.6% Living in Poverty (U.S. avg. 10.6%)

Most at Risk

1. **Children**
2. **Adults with chronic heart/lung disease**
3. **Low-income & minority populations**

Sources: Environmental Protection Agency (EPA), California Air Resources Board (CARB), U.S. Census Bureau, Federal Reserve Economic Data (FRED)

02

Data and Findings

What are we looking at?

Initial PM 10 Data

State Code	County Co	Site Num	Parameter	POC	Latitude	Longitude	Datum	Parameter	Sample Du	Pollutant S	Metric Use	Method Na	Year	Units of Me	Event Type
1	3	10	44201	1	30.49748	-87.8803	NAD83	Ozone	1 HOUR	Ozone 1-h	Daily maxir	INSTRUME	2025	Parts per n	No Events
1	3	10	44201	1	30.49748	-87.8803	NAD83	Ozone	8-HR RUN	Ozone 8-H	Daily maximum of 8 hc		2025	Parts per n	No Events
1	3	10	44201	1	30.49748	-87.8803	NAD83	Ozone	8-HR RUN	Ozone 8-H	Daily maximum of 8 hc		2025	Parts per n	No Events
1	3	10	44201	1	30.49748	-87.8803	NAD83	Ozone	8-HR RUN	Ozone 8-h	Daily maximum of 8-h		2025	Parts per n	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	1 HOUR		Observed \	Met One B/	2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 Annu	Quarterly Means of Da		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 Annu	Quarterly Means of Da		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 Annu	Quarterly Means of Da		2025	Microgram	No Events
1	3	10	88101	3	30.49748	-87.8803	NAD83	PM2.5 - Lo	24-HR BLK	PM25 Annu	Quarterly Means of Da		2025	Microgram	No Events
1	27	1	88101	3	33.28493	-85.8036	NAD83	PM2.5 - Lo	1 HOUR		Observed \	Met One B/	2025	Microgram	No Events
1	27	1	88101	3	33.28493	-85.8036	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
1	27	1	88101	3	33.28493	-85.8036	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
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1	27	1	88101	3	33.28493	-85.8036	NAD83	PM2.5 - Lo	24-HR BLK	PM25 Annu	Quarterly Means of Da		2025	Microgram	No Events
1	49	1003	88101	3	34.28857	-85.9699	NAD83	PM2.5 - Lo	1 HOUR		Observed \	Met One B/	2025	Microgram	No Events
1	49	1003	88101	3	34.28857	-85.9699	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events
1	49	1003	88101	3	34.28857	-85.9699	NAD83	PM2.5 - Lo	24-HR BLK	PM25 24-h	Daily Mean		2025	Microgram	No Events

- PM10 data was collected from the EPA Air Quality System (AQS) database
- Key PM10 variables included date, county, AQI, and particulate concentration.
- Ranges from 2016-2025

NRT VIIRS 375 m Active Fire Data

- Individual fire detection points with latitude, longitude, date, and confidence score
- Used range: 2016 - 2023

Raw data

NASA VIIRS (FIRMS) — Raw Fire Detections (sample of 7-day pull)

All confidence levels · No county label · acq_time is integer · No datetime column · All of CONUS

Keep								⚠ Needs cleaning		
latitude	longitude	bright_ti4	scan	track	acq_date	satellite	frp	confidence[1]	acq_time[2]	daynight
40.72	-92.72	342.58	0.38	0.58	2026-05-27	N	2.23	nominal	1824	D
18.35	-91.48	306.63	0.58	0.52	2026-05-23	N	1.90	nominal	822	N
20.31	-89.24	335.21	0.49	0.49	2026-05-20	N	11.76	nominal	1849	D
22.07	-105.21	354.96	0.52	0.42	2026-05-25	N	12.31	nominal	2038	D
33.63	-105.38	309.29	0.33	0.55	2026-05-23	N	1.16	nominal	820	N

[1] confidence: 'l' (low), 'n' (nominal), 'h' (high) — low-confidence rows will be dropped

[2] acq_time stored as integer (e.g. 735 = 07:35 UTC) — needs parsing into POSIXct

Source: NASA FIRMS SUOMI VIIRS C2, downloaded via FIRMS API

Cleaned data

NASA VIIRS — Cleaned Fire Detections (Imperial & Fresno)

Filtered by bounding box · Confidence ∈ {n,h} · De-duplicated · Parsed datetime

✓ New columns derived								
county[1]	datetime	year	month	date	latitude	longitude	frp	confidence
Fresno	2016-01-07 20:16 UTC	2016	1	2016-01-07	36.65141	-119.5959	1.8	n
Fresno	2016-01-07 20:16 UTC	2016	1	2016-01-07	36.65264	-119.5971	1.8	n
Fresno	2016-01-07 21:56 UTC	2016	1	2016-01-07	36.80673	-119.9110	19.7	n
Imperial	**2016-01-01 09:09 UTC**	**2016**	**1**	**2016-01-01**	**32.71898**	**-114.9616**	**0.9**	**n**
Imperial	**2016-01-01 09:09 UTC**	**2016**	**1**	**2016-01-01**	**32.93815**	**-115.8108**	**0.8**	**n**
Imperial	**2016-01-11 20:39 UTC**	**2016**	**1**	**2016-01-11**	**33.01965**	**-115.4667**	**1.8**	**n**

[1] Bold rows = Imperial County detections

Source: NASA FIRMS VIIRS C2 | Cleaned via 02_get_fire.R

CalEnviroScreen

- California Office of Environmental Health Hazard Assessment (OEHHA), 2021
- Resolution: Census tract level across all California counties

CalEnviroScreen 4.0 — Raw Data (8,035 census tracts, 66 columns)

Data is well-structured. Only issue: -999 encodes missing values instead of NA

Clean columns						⚠ -999 = missing (not a real value)		
Tract	County	TotPop19	CIscore	Asthma	Poverty	Unempl[1]	Ling_Isol	LowBirtWt
6083980100	Santa Barbara	10	-999.0	31.6	-999.0	-999.0	-999.0	-999.0
6083980000	Santa Barbara	0	-999.0	30.5	-999.0	-999.0	-999.0	-999.0
6083002915	Santa Barbara	834	8.7	30.5	27.7	-999.0	1.4	2.7
6083002103	Santa Barbara	4495	36.0	82.2	57.5	3.1	18.2	6.0
6083002402	Santa Barbara	13173	37.0	48.8	51.0	7.2	18.2	4.0
6083002102	Santa Barbara	2398	31.2	82.2	29.4	4.3	13.3	5.0

[1] -999 appears in 19 columns (103–335 tracts each). Treated as NA throughout analysis.

Source: CalEnviroScreen 4.0 Shapefile, OEHHA 2021 · 8,035 tracts · 66 columns

CalEnviroScreen 4.0 — After Cleaning (na_if(-999))

Single transformation: mutate(across(where(is.numeric), ~ na_if(.x, -999)))

Unchanged						✓ -999 → NA (R-native missing)		
Tract	County	TotPop19	CIscore	Asthma	Poverty	Unempl[1]	Ling_Isol	LowBirtWt
6083980100	Santa Barbara	10	NA	31.6	NA	NA	NA	NA
6083980000	Santa Barbara	0	NA	30.5	NA	NA	NA	NA
6083002915	Santa Barbara	834	8.7	30.5	27.7	NA	1.4	2.7
6083002103	Santa Barbara	4495	36.0	82.2	57.5	3.1	18.2	6.0
6083002402	Santa Barbara	13173	37.0	48.8	51.0	7.2	18.2	4.0
6083002102	Santa Barbara	2398	31.2	82.2	29.4	4.3	13.3	5.0

[1] Green cells were -999 in raw data. NA rows are retained; downstream functions use na.rm = TRUE.

Cleaned via: mutate(across(where(is.numeric), ~ na_if(.x, -999)))

Methodology: Dust Exposure

Data Sources:
EPA Air Quality System (AQS)
- Pulled Daily PM10 measurements

Calculations;
Monthly Climatology:
- Daily values average within each calendar month across years

Annual Mean:
- Daily values averaged within each year

Days Exceeding Threshold:
- Count of daily values

Challenges:
- Data Volume and Format
- Montor coverage of gaps of 15 CA counties thathave no PM10 monitor
- Choosing right reference threshold
- Addressing Fresno's Wildfire signal

Analysis Process:
- Analyzed similar counties
- Used reference threshold of 50 µg/m³

Health Analysis Process & Sources

CDPH California Breathing — Asthma ED Visits

California Department of Public Health · data.chhs.ca.gov

COUNTY	YEAR	STRATA	AGE GROUP	ED VISITS	RATE/10K
Imperial	2019	Total population	All ages	1,247	68.3
Imperial	2019	Age groups	0-4 years	234	142.7
Imperial	2019	Race/Ethnicity	Hispanic	1,096	71.2
Fresno	2019	Total population	All ages	6,158	60.8
Los Angeles	2019	Total population	All ages	52,041	53.2

- 58 California counties, 2015–2023
- Stratified by age and race/ethnicity
- Counts ≤10 suppressed for privacy

Asthma no significant county-level association

- bivariate β = 0.105, $p = 0.62$
- adjusted β = -0.47, $p = 0.13$

CHHS COVID-19 — Cases & Deaths by County

CA Health & Human Services · data.chhs.ca.gov

AREA	DATE	POPULATION	CUM. CASES	CUM. DEATHS
Imperial	2022-12-31	181,215	82,162	883
Los Angeles	2022-12-31	10,014,009	3,556,142	34,103
Fresno	2022-12-31	1,008,654	297,845	2,298
Kern	2022-12-31	917,673	295,124	2,284
San Diego	2022-12-31	3,338,330	964,728	5,798

- 58 California counties, daily records 2020–2023
- Multiple metrics per row (cases, deaths, tests)
- Used cumulative deaths through 2022-12-31

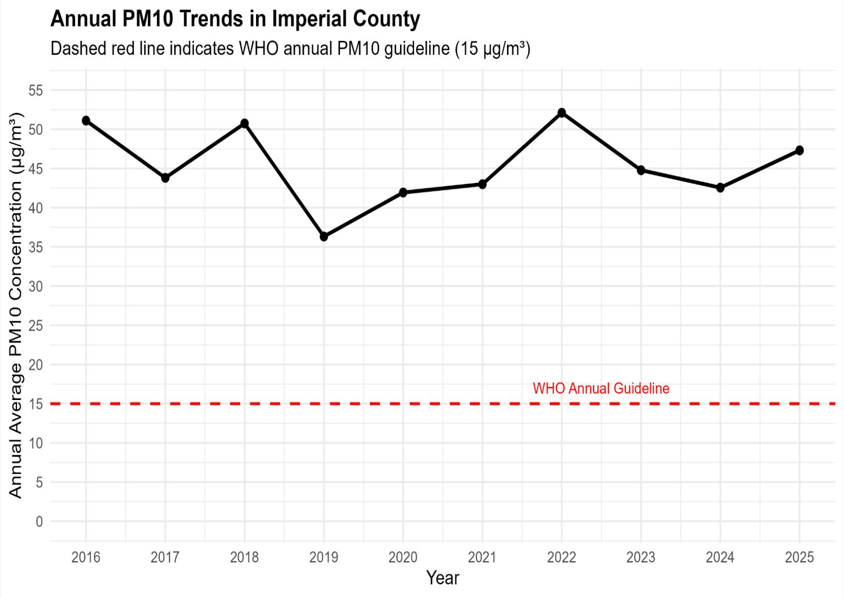
Covid: strong, robust across 43 counties

- adjusted β = -0.47, $p = 0.13$
- weighted $r = 0.59$, $p < 0.001$

Focused on COVID-19 as a health-related outcome.

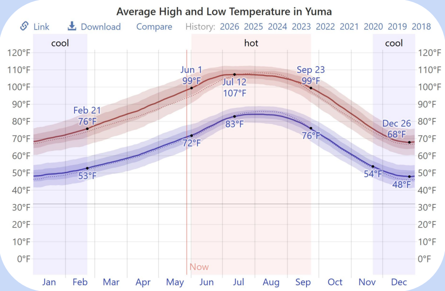
Imperial PM10 levels

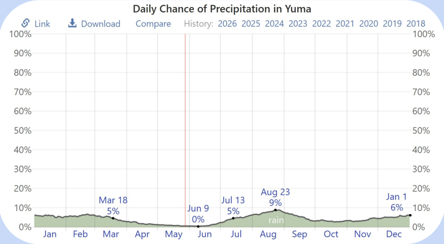
- Imperial County regularly records PM10 levels above healthier air-quality thresholds.
- Trends suggest air quality instability rather than steady improvement over time.
- Seasonal weather and drought conditions likely influence annual PM10 variation.
- Guideline is 15 grams per meter cubed for annual exposure, or 50 for daily variation exposure



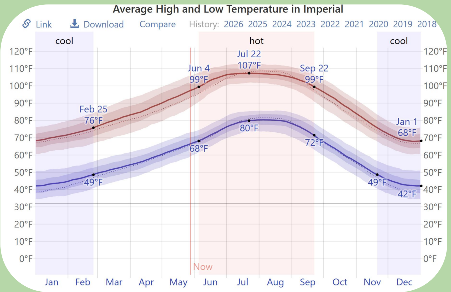
Control Groups vs Imperial (Weather)

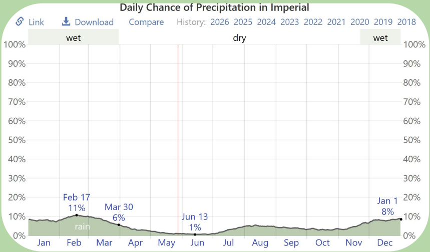
Yuma, AZ



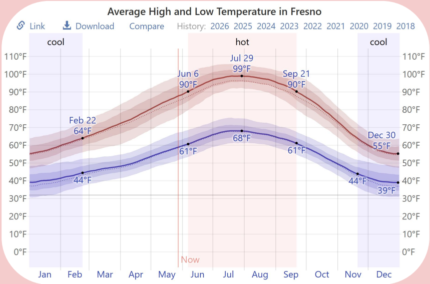


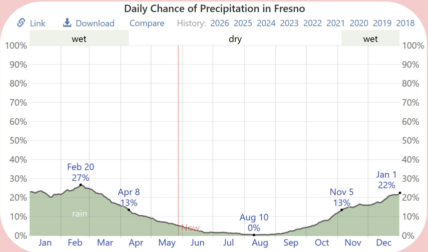
Imperial, CA





Fresno, CA





Source: Weather Spark

Control groups vs Imperial (Demographics)

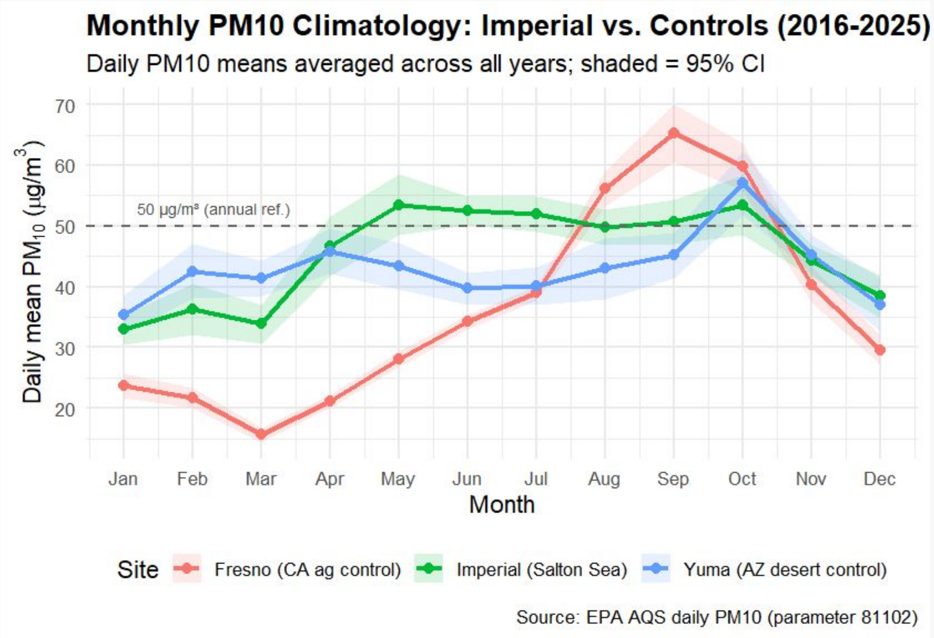
	Yuma County	Imperial County	Fresno County
Hispanic Population	66%	87%	55%
Median Household Income (U.S. med: $83,730)	$66,844	$60,749	$75,585
Living in Poverty (U.S. avg. 10.6%)	16.8%	19.6%	18.3%

Source: American Community Survey

Monthly Chronic Dust Exposure

Seasonal Swings:

- Imperial County's Wind Season March-May
- Fire Season in CA: May - November
- Imperial's PM10 levels remain high year round, while Fresno's rises during fire season
- Yuma has a similar seasonal climate to Imperial, but has lower PM10 levels

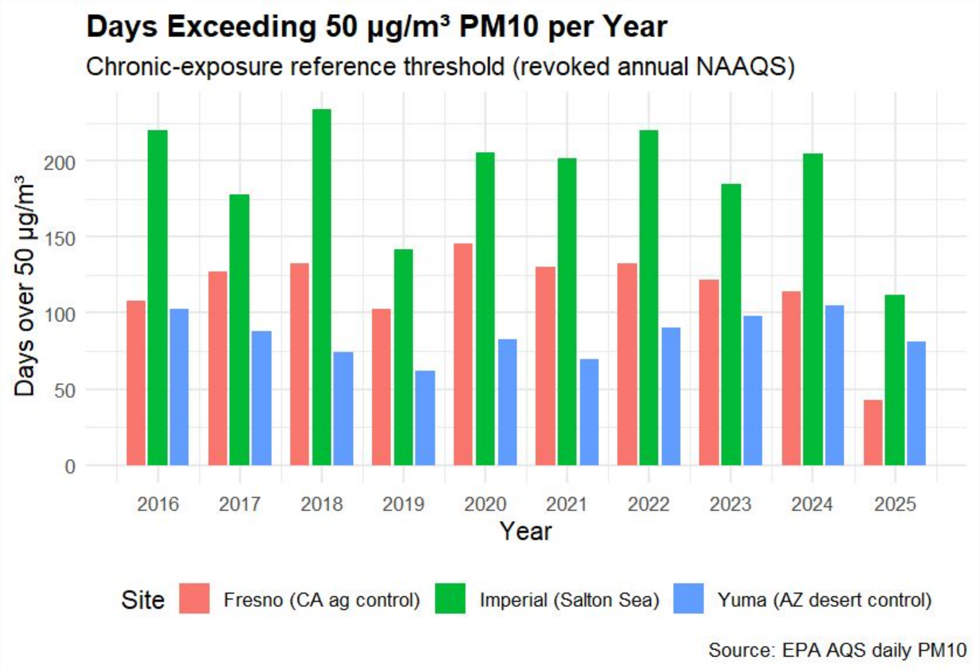


(*Fire Disaster Information*, 2024)

Days over Chronic Exposure Threshold

Consistent Daily Exposure:

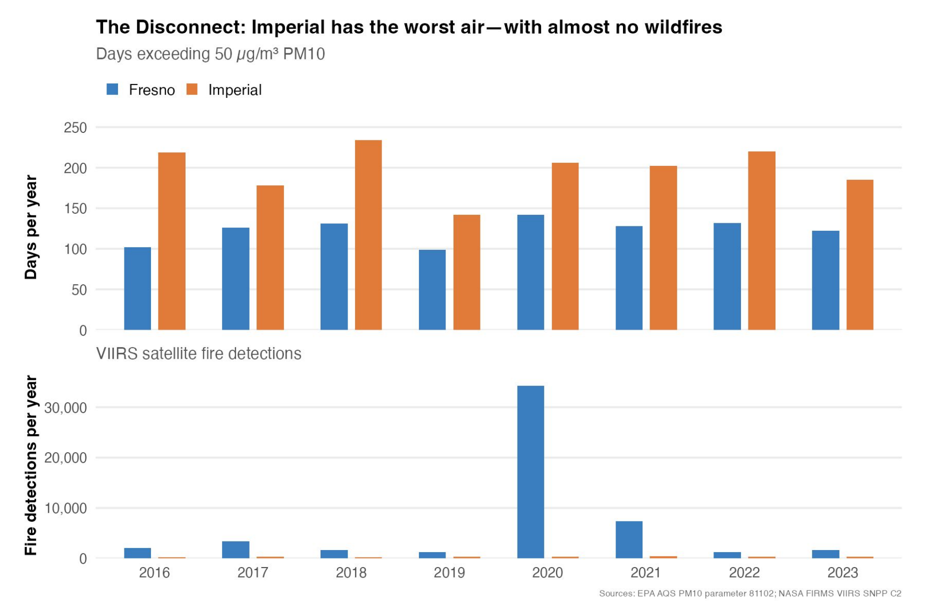
- Consistently highest number of days exposed
    - 140-230 days per year



Data with the controls: Fire

- Imperial county consistently exceeds PM10 rates compared to Fresno

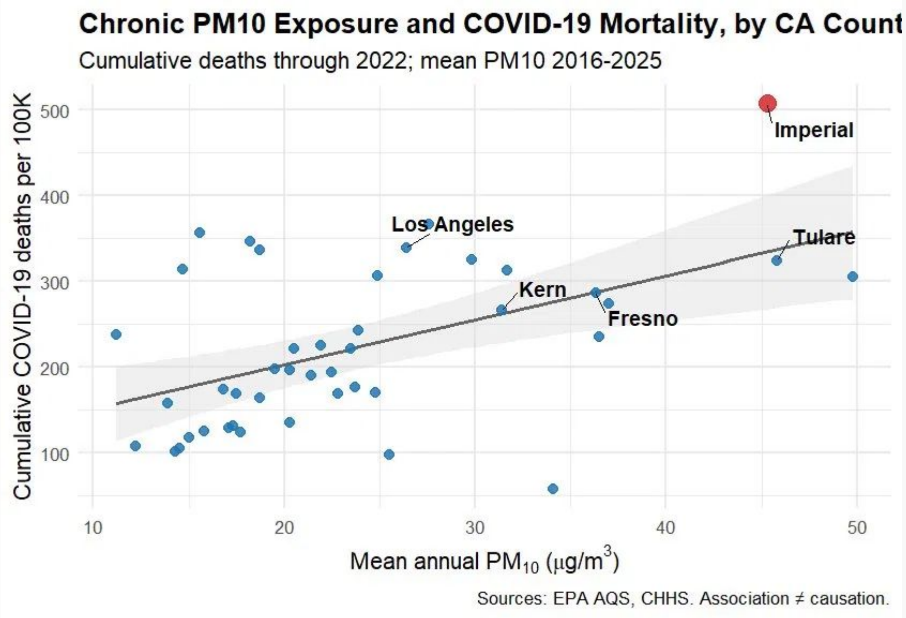
- Worst fire year for Fresno led to similar PM10 rates to those in Imperial



COVID-19 Mortality vs. Chronic PM10 Across CA

Other Plausible Explanations:

- Healthcare access (poorer counties)
- Frontline worker concentration
- Limited testing
- Housing density



Further Investigation

~3.8 Additional COVID-19 Deaths per 100k for each 1 μg/m³ increase in chronic PM10 **after adjusting for**

- Poverty
- Ethnicity
- Housing burden
- Age

COVID-19 mortality and chronic PM10 exposure across California counties

Linear regression, N = 43 counties. Outcome: cumulative COVID-19 deaths per 100K (through 2022).

PM10 COEFFICIENT: BIVARIATE VS ADJUSTED

Model	PM10 coefficient	Std. error	p-value	R^2
Bivariate (PM10 only)	5.18	1.39	0.0006	0.25
Adjusted (+ poverty, % Hispanic, housing burden, % age 65+)	**3.83**	**1.97**	**0.0596**	**0.49**

ADJUSTED MODEL: FULL COEFFICIENTS

Predictor	Estimate	Std. error	p-value
(Intercept)	-311.00	123.00	0.016 *
PM10 (μg/m³)	3.83	1.97	0.060 .
% Poverty	2.64	1.64	0.115
% Hispanic	1.37	1.08	0.211
% Housing-burdened	11.80	4.78	0.018 *
% Age 65+	7.24	4.01	0.079 .

Significance codes: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1.

PM10 effect partially attenuates after adjustment (~26% reduction) but remains positive and borderline-significant. Housing burden emerges as the strongest individual predictor of county-level COVID mortality.

Sources: EPA AQS (PM10), CHHS (COVID-19 deaths), CalEnviroScreen 4.0 (county confounders).

03

Analysis

Translating our graphs into understandable findings

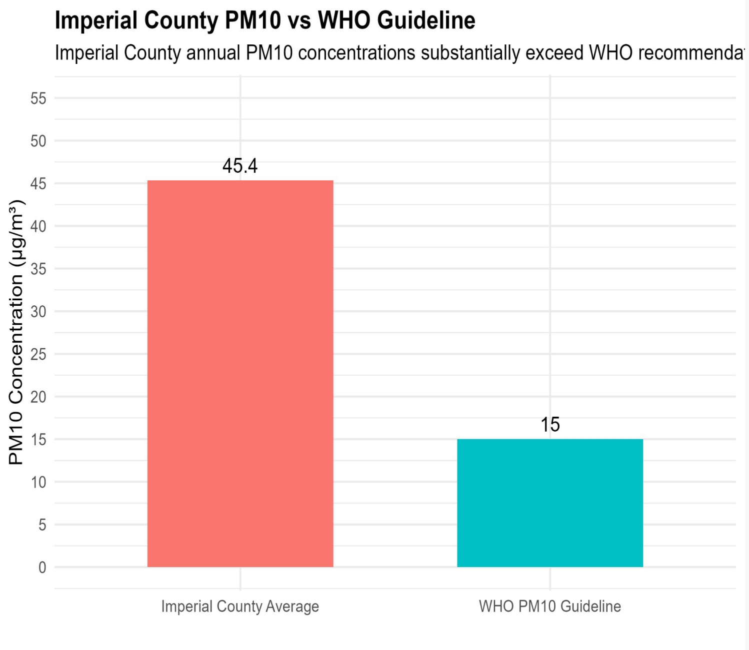
Are the Wildfires the Cause?

- 2020 Creek Fire: 34,000+ satellite detections in Fresno
- Imperial: near zero fire detections, every year
- Imperial: ~200 days/year over the 50 µg/m³ PM10 threshold
    - Fresno: ~120 days, Yuma: ~85 days (desert control)

Fires are episodic, Imperial’s dust is chronic

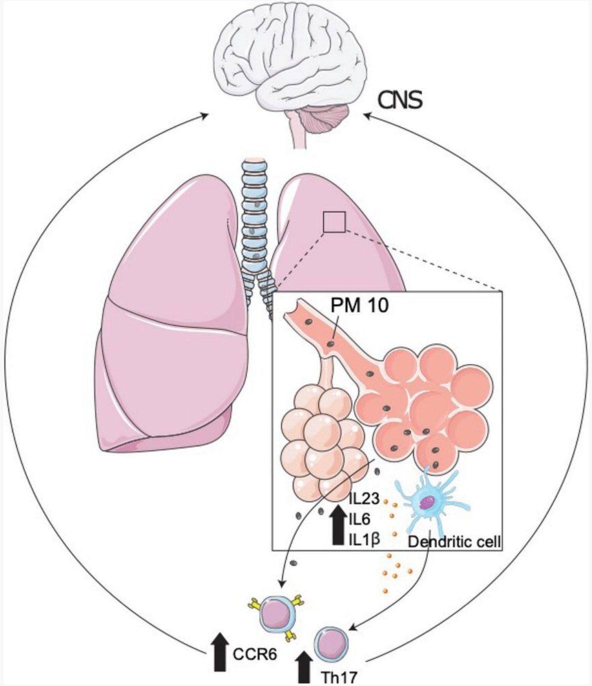
Why Imperial's elevated PM10 matters

- PM10 levels remain well above WHO recommended exposure guidelines.
- Dust pollution may contribute to increased respiratory health risks.
- Long-term exposure to particulate matter is linked to asthma and lung irritation.



Connections of Dust to Health

“PM10 is the particulate component of air pollution that can enter the lungs, deposit in the airways and also penetrate to the periphery of the lungs. There is good epidemiological evidence that asthma symptoms can be worsened by increases in PM10” (Donaldson et. al)



04

Policy Review

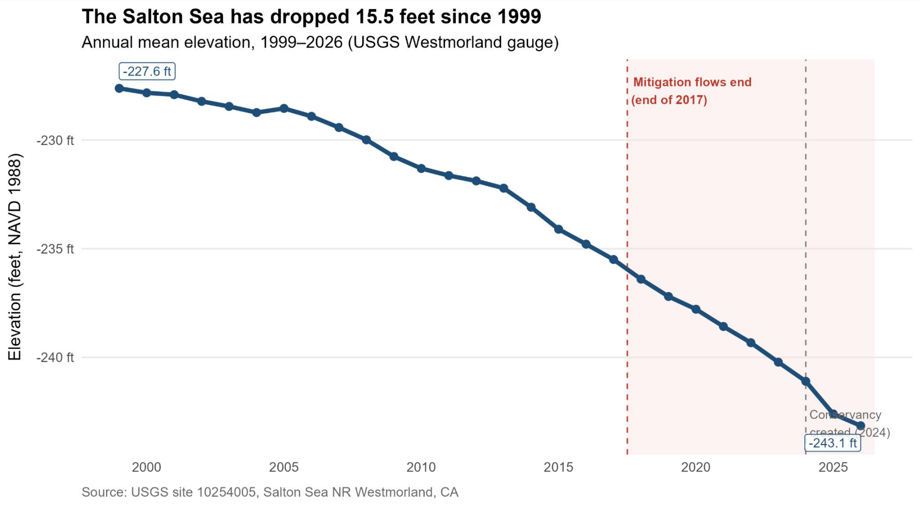
What should be done going forward

The Salton Sea Program

- SB 583 (2024) established the Salton Sea Conservancy → California's first new state conservancy in 15 years
- Approximately $500M assembled across federal, state bond, and lithium royalty sources
- First major restoration project (~9,400 acres) reached water-filling stage in 2025

Current Pace is Insufficient

- Additional playa is exposed each year that implementation lags
- Funding model depends partly on lithium revenue that has not yet materialized
- 8 year old restoration project → little to no effect on PM10 levels



USGS site 10254005 - "Salton Sea NR Westmorland, CA”

Current Pace is Insufficient

The Salton Sea is drying — faster than ever

Annual mean elevation from USGS Westmorland gauge, 1999–2026

15.5	0.50	0.84	1.7×
feet dropped since 1999	ft/year during mitigation (2003–2017)	ft/year post-mitigation (2018–today)	faster decline than during mitigation

Source: USGS site 10254005.

USGS site 10254005 - "Salton Sea NR Westmorland, CA”

Recommendations

1.

Accelerate restoration project delivery → match the pace of the drying lake

2.

Invest in Imperial County respiratory health → protect residents during the remediation window

Thanks for Listening

Any Questions?

Fires & Asthma

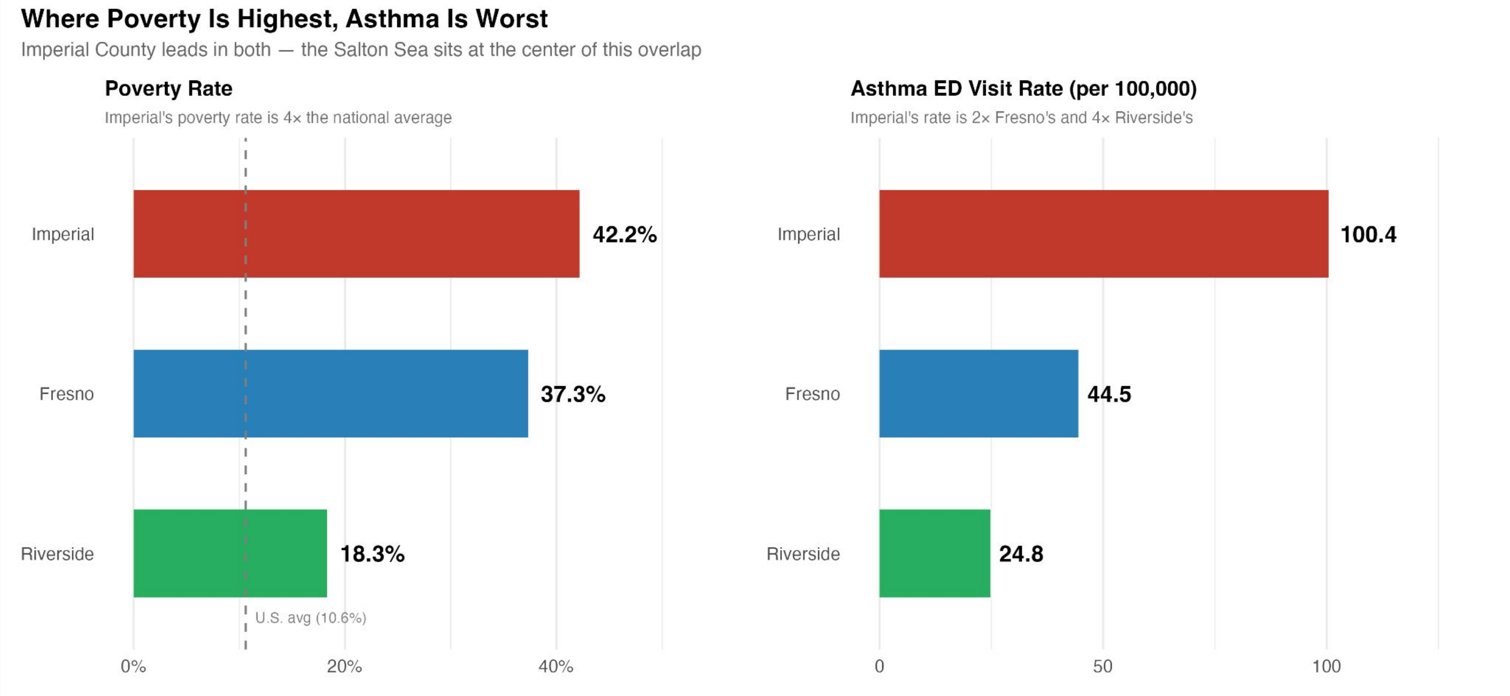
Imperial vs Fresno: PM10, Wildfire & Asthma (2016–2023)

Imperial exceeds PM10 standards *without* the wildfires

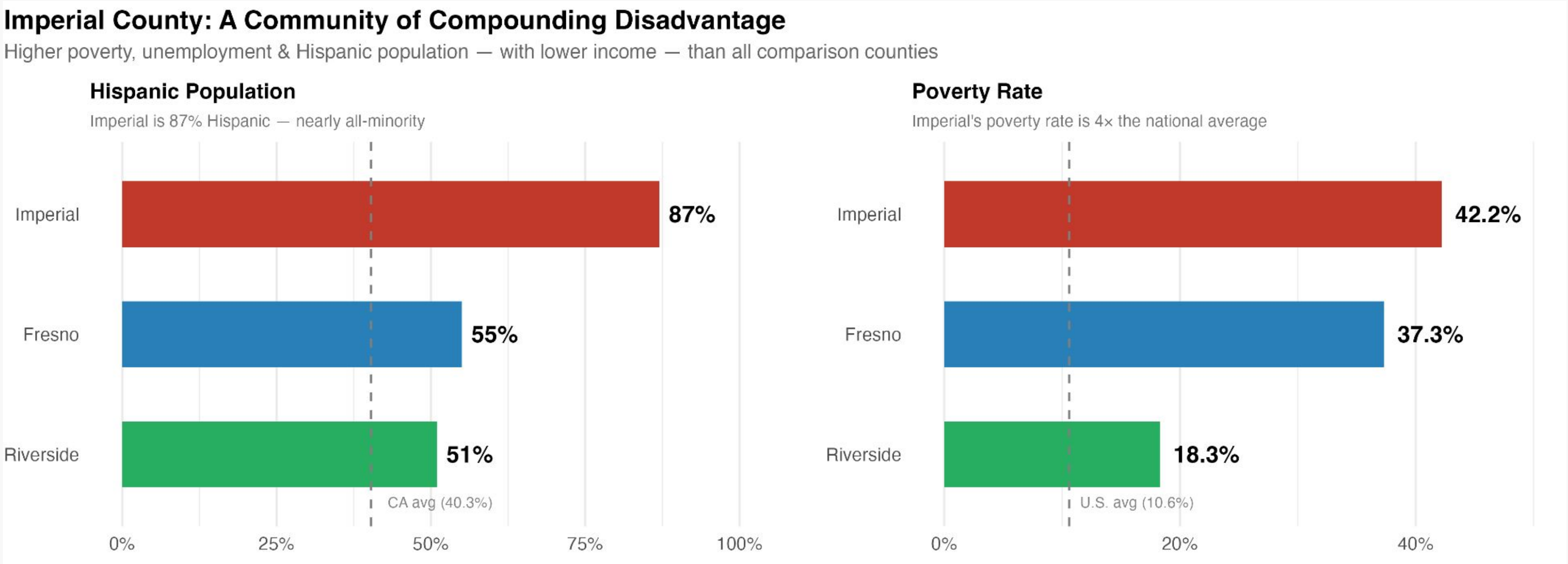
	Air Quality (PM10)			Fire Activity	Health Outcome
	Mean PM10 (µg/m³)	% Days > 50 µg/m³	Days > 150 µg/m³	Fire Detections	Asthma ED Rate (per 10k)
Fresno	38.0	23.3	309	52,828	44.2
Imperial	48.7	29.7	1037	2,392	33.4
California Avg*	43.9	26.9	1346	—	—

*CA Avg based on Imperial & Fresno monitors only (EPA AQS param 81102) | Asthma: CDPH ED visit rates (age-adjusted)

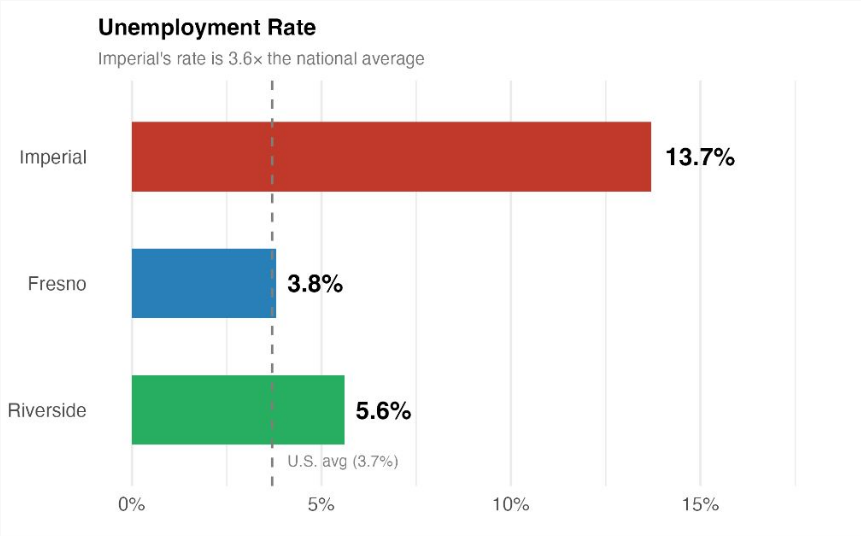
Poverty Rate and Asthma Visit



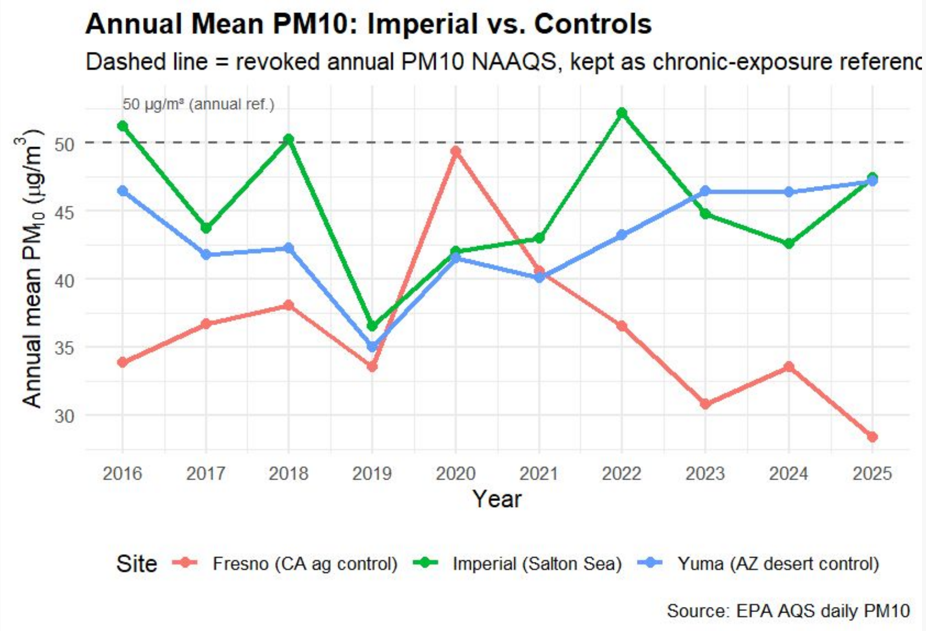
Hispanic Population



Unemployment Rate



Annual Chronic Dust Exposure



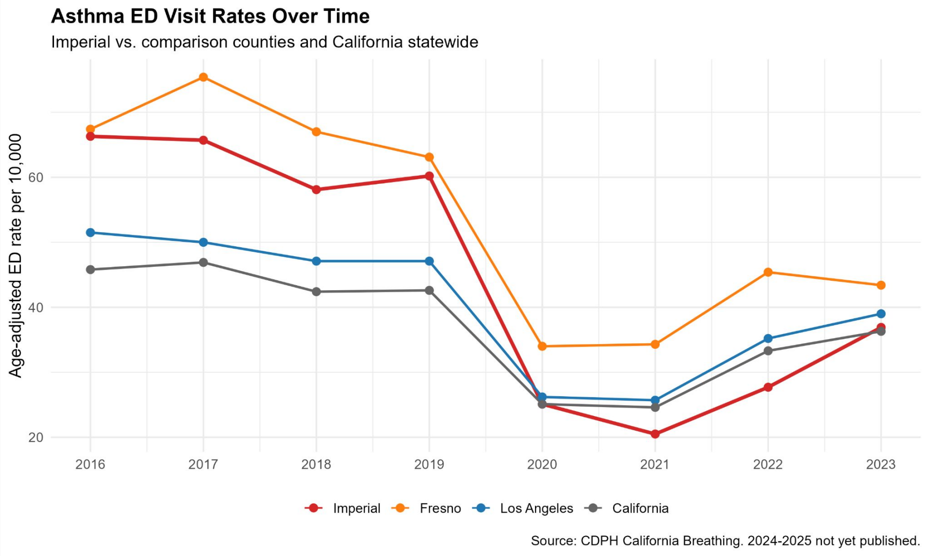
Methods for cleaning fire data

NASA VIIRS — Cleaning Decision Log

Every step documented for reproducibility

#	Cleaning Step	Raw state	Clean state	Rationale
1	**Load shapefile**	Raw SHP · mixed-case column names	Tibble (geometry dropped after bbox join)	sf needed for spatial filter; geometry not needed after
2	**Standardise column names**	LATITUDE, FRP, ACQ_TIME ...	latitude, frp, acq_time ...	Consistent with tidyverse naming convention
3	**Bounding-box filter**	All of CONUS / full archive	Imperial (bbox) + Fresno (bbox)	Focus analysis; reduce data volume ~95%
4	**Confidence filter**	confidence ∈ {l, n, h} — includes low	confidence ∈ {n, h} only	Low-confidence detections are likely false positives
5	**De-duplicate detections**	Interscan & nearby repeat detections	Unique events: lead (1km/5min) + lag (500m/1hr)	Multi-satellite passes double-count same fire event

Asthma & PM10



Asthma ED Visit Rates Over Time
Imperial vs. comparison counties and California statewide
Age-adjusted ED rate per 10,000
60
40
20
2016
2017
2018
2019
2020
2021
2022
2023
Imperial
Fresno
Los Angeles
California
Source: CDPH California Breathing. 2024-2025 not yet published.

COVID-19 & PM10